INCH-POUND
MIL-DTL-24643/28G
27 November 2023
SUPERSEDING
MIL-DTL-24643/28F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LS1S50MU (INCLUDING VARIATION LS1S50MUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS1S50MU and LS1S50MUS (double overall shield).

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated, ASTM B286, size 22 through size 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white or natural. Nominal wall thickness of 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth	-	Braided shield of 36 AWG, tin-coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth	-	Shield insulation of two polyester tapes; outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth	-	The required number of singles (see table I or table II) cabled with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to make a firm, well-rounded assembly.
Seventh	-	An optional binder.

Construction for Variant Type LS1S50MU

Eighth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Construction for Variant Type LS1S50MUS

Eighth	-	Braided shield of tin-coated copper.
Ninth	-	Spirally wrapped, 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum lap.
Tenth	-	Braided shield of tin-coated copper.
Eleventh	-	An optional binder.
Twelfth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. See table II for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (unshielded).

Military part no. M24643/28	Type and size	Number of conductors	Overall diameter (inches)	
			min.	max.
-01UN	LS1S50MU-16	16	0.760	0.825
-02UN	LS1S50MU-20	20	0.835	0.905
-03UN	LS1S50MU-40	40	1.095	1.185
-04UN	LS1S50MU-70	70	1.465	1.555

TABLE II. Details (shielded).

Military part no. M24643/28	Type and size	Number of conductors	Overall diameter (inches)	
			min.	max.
-01UD	LS1S50MUS-16	16	0.820	0.885
-02UD	LS1S50MUS-20	20	0.895	0.965
-03UD	LS1S50MUS-40	40	1.155	1.245
-04UD	LS1S50MUS-70	70	1.525	1.615

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	17.54
Voltage withstand (volts, root mean square, min.)	
Conductor to shield	5,000
Component shield to component shield	500

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Component shield to overall shield (type LS1S50MUS only)	500
Overall shield to overall shield (type LS1S50MUS only)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to shield	2,500
Component shield to component shield	100
Component shield to overall shield (type LS1S50MUS only)	100
Overall shield to overall shield (type LS1S50MUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (type LS1S50MUS only)	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (MHz) (ohms)	50±2
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Attenuation, at 400 MHz (decibel [dB]/100 feet max.)	18
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (type LS1S50MUS only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS1S50MU and LS1S50MUS (all sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-044)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.